

Extra Credit

9/21/26

the Gaussian distribution is at FWHM
when $x = \mu$.

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2}$$

we can set $\frac{1}{\sigma\sqrt{2\pi}} = A$ to simplify
the calculation

$$f(x) = A e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2}$$

since $x = \mu$, $f(\mu) = A e^0 = A$

FWHM is $f(x) = \frac{A}{2}$

we can set these equal to one another
and solve

$$A e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2} = \frac{A}{2}$$

$$e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2} = \frac{1}{2}$$

$$\left(-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2\right) = \ln(1/2) = -\ln(2)$$

$$\left(-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2\right) = -\ln(2)$$

$$\frac{(x-\mu)^2}{2\sigma^2} = \ln(2)$$

$$(x-\mu)^2 = 2\sigma^2 \ln(2)$$

$$(x-\mu) = \sigma(2\ln(2))^{1/2}$$

$$x = \mu \pm \sigma(2\ln(2))^{1/2}$$

consider x as both,

$$x = \mu + \sigma(2\ln(2))^{1/2}, \quad x = \mu - \sigma(2\ln(2))^{1/2}$$

$$x = (\mu + \sigma(2\ln(2))^{1/2}) - (\mu - \sigma(2\ln(2))^{1/2})$$

$$x = 2\sigma(2\ln(2))^{1/2}$$

$$x = \sigma 2.3548200... \approx 2.354\sigma$$

so, the value γ for FWHM is,

$$\gamma \approx 2.354\sigma$$